

Dominik Linn, Ph.D.

Senior Systems & Software Engineer | Go, C++ & High-Performance Computing

Stolberg (Aachen Area), Germany | d.linn@gmx.net | Remote & Hybrid

PROFILE

Senior Systems & Software Engineer with a Ph.D. in Technomathematics and 10+ years of experience building high-performance software. Specialist in real-time edge/cloud systems, distributed Go microservices, C++ numerical compute engines, and time-series telemetry. Combines deep mathematical rigor (SIMD, automatic differentiation, algorithmic optimization) with pragmatic domain-driven architecture, agentic developer workflows, and RFC-led technical leadership.

TECHNICAL SKILLS

Languages:	Go, C++ (11–20), Python (JAX, NumPy, SciPy), C#, SQL, Bash/Unix
Architecture & Cloud:	Hexagonal Architecture, Strategic DDD, Distributed Microservices, AWS (SQS, S3), Docker, K8s, Terraform
Edge & Telemetry:	Edge IoT Compute, Time-Series Databases (ClickHouse , Mimir/Grafana), gRPC, Protobuf, REST (OpenAPI)
Developer Productivity:	Agentic AI Workflows, Model Context Protocol (MCP), Rapid Architectural Prototyping, CLI Tooling
HPC & Numerics:	SIMD Optimization, Automatic Differentiation (Autodiff), Sparse Hessians/Jacobians, OpenMP, MPI, Eigen

PROFESSIONAL EXPERIENCE

Senior Software Engineer — [gridX](#) 2023 -- Present | Aachen / Remote, Germany

- Architect real-time Energy Management System (EMS) control engines in **Go**, applying **Hexagonal Architecture** and **Strategic DDD** to manage 200+ edge assets.
- Initiated and architected an internal **agentic AI automation tool** (custom agent skills & MCP integrations) for complex EMS bug diagnosis, significantly reducing on-call toil.
- Designed domain-agnostic tracing & explainability frameworks for edge control loops and authored foundational RFCs to drive cross-team alignment.
- Engineered edge-to-cloud telemetry pipelines using **AWS**, **ClickHouse**, and **Grafana/Mimir**, implementing local percentile aggregation for low-bandwidth environments.

Software Engineer & Applied Researcher — [Fraunhofer ITWM](#) 2018 -- 2022 | Kaiserslautern, Germany

- Architected the core district heating simulator in **C++** and **Python** for large-scale models (10^9 + DOFs), serving as lead developer for industrial client projects.
- Co-developed a C++ automatic differentiation (autodiff) engine with **Eigen** and a vectorized **JAX** solver, achieving a **100x–1000x speedup** over baseline numerical codebases.
- Parallelized compute-intensive solvers across HPC clusters using **SIMD**, **C++ Threads**, **OpenMP**, and **MPI**.
- Established modern SE practices, including **Scrum**, **GitLab CI** pipelines, and automated testing (**GoogleTest**, **Catch2**).

Software Engineer (Cross-Domain Systems) — [Fraunhofer ITWM](#) 2016 -- 2018 | Kaiserslautern, Germany

- Engineered high-performance simulation software and vector mathematical models in **C++** and **Python** across optics, energy, and biomedical robotics domains.

EDUCATION

Ph.D. (Dr. rer. nat.) in Technomathematics — TU Kaiserslautern / Fraunhofer ITWM <i>Dissertation:</i> Mathematical modeling, dynamic simulation, and optimal control of network systems	2018 – 2022
M.Sc. in Technomathematics (Minor: Physics) — TU Kaiserslautern <i>Erasmus Exchange Semester:</i> Chalmers University of Technology, Gothenburg (2012 – 2013)	2012 – 2016
B.Sc. in Mathematics (Minor: Physics) — TU Kaiserslautern	2009 – 2012